Assignment – 0			
COURSE INSIGHTS			
Q. No.		Details	Source
Q.1	1a	How many buckets are there in VIT autonomous curriculum (syllabus)?	There are 16 curriculum buckets (verticals/sub-verticals) in the VIT Autonomous Curriculum (R-2023) for the B.Tech. Computer Engineering programme.
	1b	In which bucket is this course (subject) included?	The Programming with R (CE17) course is included in the Programme Course bucket as a Programme Core Course (PC_PCC).
	1c	How many credits need to be completed from this Bucket?	44 credits are required to be completed from the Programme Core Course (PC_PCC) bucket.
	1d	How many credits from this bucket have you completed till now i.e. at the end of last semester?	Most of the Programme Core Course (PC_PCC) credits have been completed, with only a few remaining to be completed in the final year.
	1e	How many credits does this course have?	1 Credit.
Q.2	2a	Have you gone through the curriculum of your course?	Yes, I have gone through the curriculum of the course.
	2b	Is there a pre-requisite of this course?	No. There is no prerequisite for this course.
	2c	If yes, do you clearly remember the concepts of the pre-requisite course?	Not Applicable (No prerequisite course).
	2d	Pl. mention the preparations you will need to do for this course e.g. revising which earlier course/s, software installations etc..	I will need to revise basic programming concepts, statistics, and data structures, install R and RStudio, practice R syntax regularly, complete laboratory exercises on time, and explore libraries such as tidyverse, dplyr, ggplot2, Shiny, and plotly.
	2e	List the names of future course/s, if any, which build on this course i.e. this course is their Pre-requisite.	None. There are no future courses for which this course is a prerequisite.
Q.3	3a	List all the Course Outcomes (COs) of this course and mention which CO you find most useful.	• CO1: Recall the fundamental concepts, syntax, and data structures of R programming. • CO2: Explain data manipulation, visualization, and statistical concepts using R. • CO3: Apply R programming for data import, preprocessing, statistical analysis, and machine learning. • CO4: Evaluate machine learning models and optimize R programs for better performance. • CO5: Compare different machine learning algorithms and select

			suitable techniques for various datasets. • CO6: Design and implement an end-to-end data science project using R. I find CO6 to be the most useful because it focuses on developing a complete end-to-end data science project using R. It integrates data preprocessing, visualization, statistical analysis, and machine learning, providing practical skills that are highly relevant to real-world applications and future career opportunities.
	3b	How are these COs mapped to the Program Outcomes (POs)?	• PO1 – Engineering Knowledge • PO2 – Problem Analysis • PO3 – Design/Development of Solutions • PO4 – Investigation • PO5 – Modern Tool Usage • PO6 – The Engineer and Society • PO7 – Environment and Sustainability • PO12 – Life-long Learning The Course Outcomes (COs) are mapped to PO1, PO2, PO3, PO4, PO5, PO6, PO7, and PO12, ensuring the development of technical knowledge, problem-solving abilities, modern tool usage, professional responsibility, sustainability awareness, and lifelong learning skills.
Q.4	4	Where do you see the applications of this course in real life and industry?	The knowledge gained from this course can be applied in data science, machine learning, business analytics, healthcare, finance, bioinformatics, and research. R is widely used for data analysis, visualization, predictive modeling, and decision-making, making it valuable in both industry and academia.
Q.5	5a	Select one Top university outside India and mention the name of their similar course.	University: Massachusetts Institute of Technology (MIT), USA Similar Course: *Introduction to Probability and Statistics (18.05)* – Uses R for statistical computing, simulation, and data visualization.
	5b	How does their curriculum compare with our course?	Both curricula focus on R programming, statistical analysis, and data visualization. However, VIT's course places greater emphasis on practical implementation through machine learning, Shiny

			dashboards, APIs, and an end-to-end data science project, while MIT's course focuses more on probability, statistics, and theoretical foundations using R.
Q.6	6a	How many lecture sessions are planned for this course?	No lecture sessions are planned. The course consists of 30 laboratory sessions.
	6b	List the suggested Textbooks, Reference books, E-books and Research papers related to the course. Are these resources available in our library? If yes, please mention their accession numbers.	The suggested resources include R for Data Science, Discovering Statistics Using R, Learning R, R in a Nutshell, and the research paper LUCIDus: An R Package for Implementing Latent Unknown Clustering.
	6c	What resources are currently uploaded on V-refer?	The resources currently uploaded on V-Refer include the Course Lecture Notes, R for Data Science, Discovering Statistics Using R, Learning R, R in a Nutshell (Reference Book), the AAP, previous question papers, PPTs, digital content, and other course-related study materials.
	6d	What are the stipulated office hours of faculty for doubt solving sessions?	Tuesday, 3:45 PM – 4:45 PM, Venue: M209.
Q.7	7a	What is the Evaluation Metric for this Theory Course i.e. What are the weightages for ISA, MSE and ESE?	Programming with R is a practical course. Its evaluation consists of ISA – 25 marks, MSE – Not Applicable, and ESE – 25 marks (Total: 50 marks).
	7b	What does ISA comprise of?	ISA comprises continuous assessment through laboratory assignments, practical exercises, programming implementation, coding performance, and overall laboratory participation throughout the semester.
	7c	What is the evaluation rubric for ISA?	The ISA evaluation rubric is based on laboratory performance, practical assignments, programming implementation, problem-solving ability, and continuous assessment throughout the semester.
	7d	What are the dates of MSE for this course?	Not Applicable. Programming with R is a practical course and does not have an MSE.
	7e	What is the date of ESE for this course?	24 October 2026 to 5 November 2026.
Q.8	8a	Is there an allied Practical Course?	AAP
	8b	If yes, how many credits does this Practical course have?	No. Programming with R is itself a practical course; therefore, there is no separate allied practical course.
	8c	What is the Evaluation Metric for this allied Practical Course i.e. Weightages for ISA and ESE?	Not Applicable, as there is no separate allied practical course.
	8d	What are the dates of ESE exam for this allied Practical course?	Not Applicable, as there is no allied practical course.

Q. 9	9	What is the date of declaration of Result?	19 December 2026.
Q.10	10	When does the next Academic Term Start?	7 July 2026.